Two-page summary

2026-08-12 11:02 PDT

Mean moderation and covariance moderation as data-generating architectures for biomarker-treatment interactions in aggregated N-of-1 trials, and their divergent power under carryover (pmsimstats team).

Goal

Simulation is the standard tool for power and sample-size calculation in aggregated N-of-1 trials, designs in which each participant is repeatedly crossed on and off treatment and many single-patient series are pooled to validate a predictive biomarker. Any such simulation must specify how the biomarker moderates the treatment effect in its data-generating process (DGP), and that specification is a substantive scientific choice rather than an implementation detail. Each of the two established parameterizations carries its own advantages and costs. The paper's goal is to formalize both, and to show that the choice between them materially changes the estimated power loss under carryover.

Under *mean moderation*, the biomarker scales the treatment effect in the conditional mean: a deterministic, proportional relationship appropriate when the biomarker directly determines drug pharmacokinetics or pharmacodynamics (for example, kidney function setting the clearance of a renally cleared drug). Under *covariance moderation*, the interaction is instead represented as a correlation between biomarker and biological response that depends on drug-exposure condition (on drug versus off drug): a probabilistic relationship appropriate when the biomarker is only an imperfect proxy for a latent responder subtype. Covariance moderation is often the more biologically defensible choice in that latter setting, and it is the specification Hendrickson et al. (2020) selected for the paper's motivating application, a prazosin trial for PTSD in which resting blood pressure is thought to index noradrenergic tone rather than directly drive drug levels. That defensibility, the paper argues, does not come free: because the covariance-moderation signal lives in a correlation that is sustained only while the drug is active, any off-drug erosion of drug effect directly erodes the statistical signal itself.

Method

Both architectures are embedded in an otherwise identical Monte Carlo simulation, holding the analysis model fixed at a linear mixed model with continuous-time AR(1) residual correlation. Power to detect the biomarker-by-treatment interaction is estimated as the carryover half-life increases from zero to one week, across three within-subject designs matched to the prazosin and PTSD reference application: a classical crossover, a Hybrid N-of-1 design, and an open-label design followed by blinded discontinuation (OL+BDC). Sample size is $N = 70$ throughout, matched across designs so the comparison is a design comparison, not a sample-size one. The study follows the ADEMP reporting framework, with 1,000 replicates per cell and every performance measure reported with its Monte Carlo standard error.

Results

Under mean moderation, power is nearly preserved as carryover increases, because carryover does not disrupt the biomarker's proportional relationship to the treatment effect: in the OL+BDC design, power falls only from 0.751 to 0.730 (a 2.8% relative loss), and losses across all three designs range from 1.9% to 6.9%. Under covariance moderation, the same carryover erodes the differential correlation that carries the signal, and power falls much further, from 0.777 to 0.539 in OL+BDC (a 30.6% relative loss), roughly eleven times the matched mean-moderation loss and far beyond Monte Carlo error ($z = 7.64$, $p < 10^{-13}$).

The gap between the two architectures is strongly design-dependent. It is largest in designs with a long blinded-discontinuation phase (OL+BDC, then Hybrid at 15.4% loss) and negligible in the classical crossover (3.0%, not statistically distinguishable from zero), because the within-subject AR(1) correlation in a crossover design already carries most of the signal that covariance moderation depends on, leaving carryover little left to erode. A robustness check at $N = 140$ on the OL+BDC design confirms that this ordering survives a larger sample, though absolute magnitudes are compressed by ceiling saturation. Type I error across all null cells sat uniformly below the nominal 5% level for both architectures alike, a known conservatism of the working covariance structure that does not disturb the architecture comparison.

Conclusion

The DGP architecture is a substantive scientific assumption, not a parameterization convenience, and it determines how severely carryover appears to reduce power. The choice should be guided by the hypothesized biological mechanism: mean moderation for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics, covariance moderation for biomarkers that statistically predict treatment response through latent subtype membership. The existing clinical trial simulation literature uses mean moderation almost exclusively, so published power estimates for biomarkers that actually operate through a correlation-based mechanism may be systematically optimistic. Biomarker-validation simulation studies should state which architecture they adopt and report power under both.